

Equilibrium of refined learning threshold with Coxian service times setting

ChatGPT (gpt-5.6-sol)*

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Abstract

We consider a system of n infinite-server pools operating under a threshold-based dispatching policy together with the refined threshold-learning scheme. Arrivals occur at rate $n\lambda$, and service requirements follow a Coxian distribution. Previous results provide compact-horizon occupancy bounds and show that the refined threshold is eventually no larger than $\lfloor \rho \rfloor_\Delta$, where ρ is the offered load per pool, but they do not establish equality of the threshold in the Coxian setting.

We supply the missing lower-bound and lock-in argument. Under the natural scaling

$$\alpha_n \longrightarrow 1, \quad \liminf_{n \rightarrow \infty} n^{\gamma_0} (1 - \alpha_n) > 0 \quad \text{for some } \gamma_0 \in (0, 1/2),$$

the refined threshold equals $\lfloor \rho \rfloor_\Delta$ throughout every fixed compact interval after a deterministic transient, for all sufficiently large n , almost surely. In addition, the fraction of missing slots below the limiting threshold is $o(n^{-\gamma})$, uniformly on compact time intervals, for every $\gamma < 1/2$. We also explain why this compact-horizon stabilization cannot be strengthened to uniform stabilization over the infinite time horizon: in every fixed finite system, the refined threshold changes infinitely often almost surely.

1 The open problem

This paper addresses an open problem posed by Diego Goldszajn, Sem Borst, Johan S. H. van Leeuwaarden in *Learning and Balancing Unknown Loads in Large-Scale Systems* [1] (Mathematics of Operations Research, 2024), Page 21, Section 4.3 (Extensions). The website catalogue entry is problem 118 (w4396625954_p0) of the [Open Problems in Operations Research](#) collection.

*Generated by the `base_solver_non_interactive` (MOR sweep) pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve $\rightarrow$ A.8 verify $\rightarrow$ A.9 write-up; run `run_20260826_212409_b614b566`, `sweep-audit-approved`). Independent verifier assessment: APPROVED with progress score 3/3 (full solution). Maintained by the AI in Mathematics & Operations Research project.

Establish conditions under which, for Coxian service times with constant arrival rate $n\lambda$, the threshold process $\ell_n(t)$ generated by the refined learning scheme converges (in the sense that for some deterministic $\ell^ \in \{0, \Delta, 2\Delta, \dots\}$ and some $\sigma > 0$, $\ell_n(t) = \ell^*$ for all $t \geq \sigma$ and all sufficiently large n with probability one) to an equilibrium value, and characterize ℓ^* in terms of ρ and Δ .*

The remainder of this document is the solution produced by the model, reproduced verbatim from the pipeline output.

2 Introduction

We consider the load-balancing model of Goldsztajn, Borst, and van Leeuwaarden [2]. Theorem 3 of that work establishes compact-horizon stabilization for Coxian service requirements when the basic learning scheme is used, and its Section 4.3 observes that, under the refined learning scheme, the occupancy conclusions and the eventual upper bound

$$\ell_n \leq \lfloor \rho \rfloor_\Delta$$

can be recovered by adapting the existing arguments. The remaining difficulty is to rule out persistent downward excursions of the threshold.

The proof below is based on a vacancy process. Set

$$j \doteq \lfloor \rho \rfloor_\Delta.$$

Once $\ell_n \leq j$, define H_n to be the normalized number of empty slots in the first j columns of the occupancy diagram. During any positive excursion of H_n , every arrival fills one such slot; in contrast, only departures from pools with pre-departure occupancy at most j can create a vacancy.

The aggregate Coxian phase equations imply that, after a deterministic transient, the rate at which vacancies are filled exceeds the rate at which they are created by a fixed positive amount. This forces H_n to hit zero, and at that time the refined update rule places the threshold exactly at j . Subsequent positive vacancy excursions start at height $1/n$. Their maximal height is $o(n^{-\gamma})$ for every $\gamma < 1/2$, which is too small to trigger a downward update under the assumed scaling of α_n .

All assertions are compact-horizon statements. More precisely, the relevant order of quantifiers is

$$\forall T < \infty \quad \exists N_T < \infty,$$

rather than a single N that works simultaneously for all future time.

3 Model and notation

3.1 Coxian service requirements

There are n statistically identical infinite-server pools. Tasks arrive according to a Poisson process A_n of rate $n\lambda$, where $\lambda > 0$ is constant.

Every task starts in phase 1 and passes through at most r phases. A phase- m task completes its current phase at rate $\mu_m > 0$; upon completing phase m , it proceeds to phase $m + 1$ with probability p_m and leaves the system with probability $1 - p_m$. We assume

$$p_m \in (0, 1) \quad (m < r), \quad p_r = 0.$$

Let

$$r_1 = 1, \quad r_m = \prod_{k=1}^{m-1} p_k, \quad 2 \leq m \leq r,$$

so that r_m is the probability that a task reaches phase m . The mean service requirement is

$$\frac{1}{\mu} = \sum_{m=1}^r \frac{r_m}{\mu_m},$$

and the offered load per pool is

$$\rho = \frac{\lambda}{\mu} = \lambda \sum_{m=1}^r \frac{r_m}{\mu_m}.$$

Write

$$a_m = (1 - p_m)\mu_m, \quad \eta = \min_{1 \leq m \leq r} a_m > 0.$$

Thus every task, regardless of its current phase, leaves the system at instantaneous rate at least η .

For $\nu = (\nu_1, \dots, \nu_r) \in \mathbb{N}^r$, let

$$|\nu| = \sum_{m=1}^r \nu_m.$$

The empirical profile distribution is

$$s_n(t, \nu) = \frac{1}{n} \# \{\text{pools with profile } \nu \text{ at time } t\},$$

and the fraction of pools containing at least i tasks is

$$q_n(t, i) = \sum_{|\nu| \geq i} s_n(t, \nu), \quad i \in \mathbb{N}.$$

In particular, $q_n(t, 0) = 1$.

The normalized phase totals and total occupancy are

$$U_{n,m}(t) = \sum_{\nu \in \mathbb{N}^r} \nu_m s_n(t, \nu), \quad u_n(t) = \sum_{m=1}^r U_{n,m}(t) = \sum_{i \geq 1} q_n(t, i),$$

and the normalized instantaneous departure rate from the system is

$$d_n(t) = \sum_{\nu \in \mathbb{N}^r} \sum_{m=1}^r a_m \nu_m s_n(t, \nu) = \sum_{m=1}^r a_m U_{n,m}(t).$$

We impose the usual nondegeneracy assumptions on the initial conditions: for some deterministic $B < \infty$,

$$q_n(0, B + 1) = 0, \quad \ell_n(0) \leq B$$

almost surely for every n , and $s_n(0)$ converges almost surely, in the product topology, to a deterministic profile distribution $s(0)$. The bounded-support assumption implies convergence of the corresponding initial phase totals.

3.2 Dispatching rule

Fix an integer $\Delta \geq 1$. The threshold $\ell_n(t)$ takes values in $\Delta\mathbb{N}$, and

$$h_n(t) \doteq \ell_n(t) + \Delta.$$

An arriving task is dispatched according to the following rule, evaluated on the pre-arrival state:

1. If some pool has fewer than ℓ_n tasks, choose uniformly at random among those pools.
2. Otherwise, if some pool has fewer than h_n tasks, choose uniformly at random among the pools having at least ℓ_n and fewer than h_n tasks.
3. Otherwise, choose uniformly at random among the pools having at least h_n tasks.

3.3 Refined learning scheme

The update conditions are evaluated immediately before an arrival, using the pre-arrival occupancy state and threshold. The arriving task is first dispatched, and the threshold is then updated by

$$\ell_n(t) = \begin{cases} \ell_n(t^-) - \Delta, & \ell_n(t^-) > 0 \text{ and } q_n(t^-, \ell_n(t^-)) \leq \alpha_n, \\ \ell_n(t^-) + \Delta, & q_n(t^-, \ell_n(t^-)) = 1 \text{ and } q_n(t^-, h_n(t^-)) \geq 1 - \frac{1}{n}, \\ \ell_n(t^-), & \text{otherwise.} \end{cases} \quad (3.1)$$

The two nontrivial cases are mutually exclusive because $\alpha_n < 1$.

We assume that, for some $\gamma_0 \in (0, 1/2)$,

$$\alpha_n \longrightarrow 1, \quad \liminf_{n \rightarrow \infty} n^{\gamma_0} (1 - \alpha_n) > 0. \quad (3.2)$$

In particular,

$$\alpha_n < 1 - \frac{1}{n} \quad (3.3)$$

for all sufficiently large n .

Finally, define

$$\lfloor x \rfloor_\Delta \doteq \max\{k\Delta : k \in \mathbb{N}, k\Delta \leq x\}.$$

4 Preliminary estimates

4.1 Aggregate phase dynamics

Routing does not affect the total number of tasks in any phase. This observation remains valid for the refined policy even though the threshold and the detailed occupancy state are mutually dependent.

Lemma 1 (Phase-total law of large numbers). *On a common probability space, for every finite T ,*

$$\max_{1 \leq m \leq r} \sup_{0 \leq t \leq T} |U_{n,m}(t) - U_m(t)| \longrightarrow 0 \quad \text{almost surely,} \quad (4.1)$$

where $U = (U_1, \dots, U_r)$ is the unique solution of

$$\dot{U}_1(t) = \lambda - \mu_1 U_1(t), \quad (4.2)$$

$$\dot{U}_m(t) = p_{m-1} \mu_{m-1} U_{m-1}(t) - \mu_m U_m(t), \quad 2 \leq m \leq r. \quad (4.3)$$

Moreover,

$$U_m(t) \longrightarrow U_m^* \doteq \frac{\lambda r_m}{\mu_m}, \quad u(t) \doteq \sum_{m=1}^r U_m(t) \longrightarrow \rho, \quad (4.4)$$

and

$$d(t) \doteq \sum_{m=1}^r a_m U_m(t) \longrightarrow \lambda. \quad (4.5)$$

In addition, $u_n \rightarrow u$ and $d_n \rightarrow d$ uniformly on every compact time interval.

Proof. Represent phase progressions and external departures by independent time-changed unit-rate Poisson processes. If $P_{n,m}$ denotes the number of phase- m progressions and $D_{n,m}$ the number of external departures from phase m , their compensators are

$$n \int_0^t p_m \mu_m U_{n,m}(s) ds \quad \text{and} \quad n \int_0^t a_m U_{n,m}(s) ds,$$

respectively. Consequently,

$$\begin{aligned} U_{n,1}(t) &= U_{n,1}(0) + \frac{A_n(t)}{n} - \frac{P_{n,1}(t)}{n} - \frac{D_{n,1}(t)}{n}, \\ U_{n,m}(t) &= U_{n,m}(0) + \frac{P_{n,m-1}(t)}{n} - \frac{P_{n,m}(t)}{n} - \frac{D_{n,m}(t)}{n}, \quad m \geq 2. \end{aligned}$$

Uniform strong laws for the Poisson primitives show that every subsequential limit satisfies (4.2)–(4.3). The linear system has a unique solution, so the full sequence converges.

The system matrix is lower triangular with diagonal entries $-\mu_1, \dots, -\mu_r$, and is therefore exponentially stable. Its equilibrium satisfies

$$U_1^* = \frac{\lambda}{\mu_1}, \quad U_m^* = \frac{p_{m-1} \mu_{m-1} U_{m-1}^*}{\mu_m} = \frac{\lambda r_m}{\mu_m}.$$

Thus

$$\sum_{m=1}^r U_m^* = \lambda \sum_{m=1}^r \frac{r_m}{\mu_m} = \rho.$$

Finally,

$$\sum_{m=1}^r (1 - p_m) r_m = \sum_{m=1}^r (r_m - r_{m+1}) = 1,$$

where $r_{r+1} \doteq 0$. Hence

$$\sum_{m=1}^r a_m U_m^* = \lambda \sum_{m=1}^r (1 - p_m) r_m = \lambda.$$

This proves (4.5) and the remaining uniform convergence assertions. $\square$

4.2 The upper-band invariant

Lemma 2 (Eventual invariant and compact-horizon bound). *Fix $0 < \delta < T < \infty$. On a probability-one event, for every sufficiently large n ,*

$$q_n(t, h_n(t)) < 1 \quad \text{for every } t \in [\delta, T]. \quad (4.6)$$

Moreover, for each fixed T , there is a deterministic $B_T < \infty$ such that, for all sufficiently large n , no pool contains more than B_T tasks during $[0, T]$, and ℓ_n is bounded there by B_T .

Proof. First note that (3.2) implies (3.3) for all sufficiently large n . Suppose that $q_n(h_n) < 1$ immediately before an arrival.

If a downward update occurs, then the new upper threshold is the old lower threshold. Since one arrival can increase $q_n(\ell_n)$ by at most $1/n$,

$$q_n(t, h_n(t)) \leq \alpha_n + \frac{1}{n} < 1.$$

If an upward update occurs, the task is routed to a pool below the old upper threshold, and that pool remains below the new upper threshold, so (4.6) is again preserved. If there is no update, then either the arrival is routed below ℓ_n , in which case $q_n(h_n)$ is unchanged, or it is routed below h_n . In the latter case, the absence of an upward update implies

$$q_n(t^-, h_n(t^-)) \leq 1 - \frac{2}{n},$$

so the arrival cannot make the upper band full. Departures also preserve (4.6).

It remains to show that the invariant is reached. While $q_n(h_n) = 1$, the upward-update condition holds at every arrival, so the threshold increases by Δ at successive arrivals. Since the initial occupancies and the initial threshold are bounded by B , this cannot continue for more than a deterministic number of arrivals when n is large: after $B + 1$ arrivals, at least one pool has not yet received a task, whereas the upper threshold would already exceed B . The arrival strong law supplies at least $B + 1$ arrivals before any fixed $\delta > 0$, for all sufficiently large n . This proves (4.6).

For the compact-horizon bound, observe first that

$$u_n(t) \leq B + \frac{A_n(T)}{n}, \quad 0 \leq t \leq T,$$

which is eventually bounded by a deterministic constant. At an upward update occurring after (4.6) has been reached,

$$u_n(t^-) = \sum_{i \geq 1} q_n(t^-, i) \geq \ell_n(t^-) + \Delta q_n(t^-, h_n(t^-)) \geq h_n(t^-) - \frac{\Delta}{n}.$$

Hence the new lower threshold, which equals the old upper threshold, is bounded by $u_n + \Delta/n$. Between upward updates the threshold cannot increase. Finally, under (4.6), every arrival is routed to a pool below the current upper threshold, so the threshold bound also yields a uniform bound on individual pool occupancies. $\square$

4.3 Strong approximations for vacancy excursions

Fix $j \in \mathbb{N}$. Define

$$x_{n,j}(t) \doteq \sum_{i=1}^j q_n(t, i) = \sum_{\nu \in \mathbb{N}^r} \min\{|\nu|, j\} s_n(t, \nu)$$

and

$$H_{n,j}(t) \doteq j - x_{n,j}(t) = \sum_{i=1}^j (1 - q_n(t, i)). \quad (4.7)$$

Thus $nH_{n,j}$ is the number of vacant slots among the first j slots across all pools.

Let $C_{n,j}(t)$ count the external departures up to time t from pools whose occupancy immediately before the departure is at most j . Its normalized predictable rate is

$$b_{n,j}(t) \doteq \sum_{|\nu| \leq j} \sum_{m=1}^r a_m \nu_m s_n(t, \nu). \quad (4.8)$$

Lemma 3 (Uniform Poisson approximation). *For every fixed $T < \infty$ and every $\gamma < 1/2$,*

$$\sup_{0 \leq s \leq t \leq T} \left| \frac{A_n(t) - A_n(s)}{n} - \lambda(t - s) \right| = o(n^{-\gamma}), \quad (4.9)$$

$$\sup_{0 \leq s \leq t \leq T} \left| \frac{C_{n,j}(t) - C_{n,j}(s)}{n} - \int_s^t b_{n,j}(v) dv \right| = o(n^{-\gamma}) \quad (4.10)$$

almost surely.

Proof. The first assertion follows from the functional law of the iterated logarithm for a unit-rate Poisson process after the time change $t \mapsto n\lambda t$.

For the second assertion, $C_{n,j}$ is a sum of time-changed Poisson primitives indexed by

$$\{(\nu, m) : |\nu| \leq j, 1 \leq m \leq r\}.$$

This is a finite set, and every corresponding time change is bounded by a constant times nT . Applying the same Poisson estimate to each primitive and summing gives (4.10). Since the suprema range over all pairs s, t , the estimates also hold when the endpoints are random excursion times. $\square$

We henceforth work on a single probability-one event on which the preceding lemmas hold for all integer horizons and all rational $\gamma < 1/2$, as well as for the exponent γ_0 in (3.2). This event is obtained by a countable intersection. For an arbitrary real $\gamma < 1/2$, one may choose a rational $\bar{\gamma} \in (\gamma, 1/2)$.

5 Compact-horizon stabilization

Assume from now on that

$$\rho \notin \{k\Delta : k \in \mathbb{N}\},$$

and put

$$j \doteq \lfloor \rho \rfloor_{\Delta}, \quad \text{so that} \quad j < \rho < j + \Delta.$$

Theorem 1 (Refined Coxian stabilization). *Under the assumptions above, there exist deterministic constants $\sigma, c > 0$ such that, on a single probability-one event, the following statements hold for every finite $T > \sigma$.*

First, for all sufficiently large n , where the required lower bound on n may depend on T ,

$$\ell_n(t) = j \quad \text{for every } t \in [\sigma, T]. \quad (5.1)$$

Second, for every $\gamma < 1/2$,

$$\lim_{n \rightarrow \infty} n^\gamma \sup_{t \in [\sigma, T]} H_{n,j}(t) = 0. \quad (5.2)$$

Consequently, for every $i \leq j$,

$$\lim_{n \rightarrow \infty} n^\gamma \sup_{t \in [\sigma, T]} (1 - q_n(t, i)) = 0. \quad (5.3)$$

Finally, with

$$Z_n(t) = \sum_{i > j + \Delta} q_n(t, i),$$

one may choose c independently of T so that

$$\limsup_{n \rightarrow \infty} \sup_{t \in [\sigma, T]} Z_n(t) e^{\eta(t - \sigma)} \leq c. \quad (5.4)$$

Proof. Choose $\varepsilon > 0$ so small that

$$5\varepsilon < \min\{\rho - j, j + \Delta - \rho\}. \quad (5.5)$$

By Lemma 1, there exists a deterministic $t_0 > 0$ such that, for every fixed $T > t_0$ and all sufficiently large n ,

$$j + 3\varepsilon \leq u_n(t) \leq j + \Delta - 3\varepsilon, \quad t \in [t_0, T], \quad (5.6)$$

$$d_n(t) \leq \lambda + \eta\varepsilon, \quad t \in [t_0, T]. \quad (5.7)$$

Increase t_0 , if necessary, so that the invariant

$$q_n(t, h_n(t)) < 1 \quad (5.8)$$

also holds throughout $[t_0, T]$ for all sufficiently large n .

Step 1: the threshold becomes and remains no larger than j .

If $\ell_n(t) > j$, then $\ell_n(t) \geq j + \Delta$, and

$$q_n(t, \ell_n(t)) \leq \frac{u_n(t)}{\ell_n(t)} \leq \frac{j + \Delta - 3\varepsilon}{j + \Delta} < 1.$$

Since $\alpha_n \rightarrow 1$, the right-hand side is smaller than α_n for all large n . Hence every arrival occurring while $\ell_n > j$ triggers a downward update.

Lemma 2 gives a deterministic compact-horizon bound on the threshold. The arrival strong law gives at least $c_0 n$ arrivals during $[t_0, t_0 + 1]$, for some $c_0 > 0$ and all sufficiently large n . In particular, there are enough arrivals to eliminate all possible multiples of Δ above j . Thus, with

$$\tau = t_0 + 1,$$

we have

$$\ell_n(\tau) \leq j.$$

An upward update from j is impossible. Indeed, its pre-arrival conditions would imply

$$\begin{aligned} u_n(t^-) &= \sum_{i \geq 1} q_n(t^-, i) \\ &\geq j + \sum_{i=j+1}^{j+\Delta} q_n(t^-, i) \\ &\geq j + \Delta q_n(t^-, j + \Delta) \geq j + \Delta - \frac{\Delta}{n}, \end{aligned}$$

contradicting the upper bound in (5.6) for all large n . An upward update from a threshold strictly below j cannot overshoot j , since all threshold values are multiples of Δ . Therefore,

$$\ell_n(t) \leq j \quad \text{for every } t \in [\tau, T]. \quad (5.9)$$

Step 2: a strict vacancy-rate gap.

The following identity is immediate from the definitions:

$$u_n(t) - x_{n,j}(t) = \sum_{\nu \in \mathbb{N}^r} (|\nu| - j)^+ s_n(t, \nu). \quad (5.10)$$

Moreover,

$$\begin{aligned} d_n(t) - b_{n,j}(t) &= \sum_{|\nu| > j} \sum_{m=1}^r a_m \nu_m s_n(t, \nu) \\ &\geq \eta \sum_{|\nu| > j} |\nu| s_n(t, \nu) \\ &\geq \eta (u_n(t) - x_{n,j}(t)). \end{aligned} \quad (5.11)$$

When $H_{n,j}(t) > 0$, we have

$$x_{n,j}(t) = j - H_{n,j}(t) < j.$$

Combining (5.6), (5.7), and (5.11) gives

$$\begin{aligned} b_{n,j}(t) &\leq d_n(t) - \eta (u_n(t) - x_{n,j}(t)) \\ &\leq \lambda + \eta\varepsilon - 3\eta\varepsilon = \lambda - 2\eta\varepsilon. \end{aligned} \quad (5.12)$$

Step 3: vacancy excursions and capture at zero.

While

$$H_{n,j} > 0, \quad \ell_n \leq j, \quad q_n(h_n) < 1,$$

every arrival is sent to a pool containing fewer than j tasks. Indeed, if $\ell_n = j$, the positivity of $H_{n,j}$ means that the first dispatching case applies; if $\ell_n < j$, then $\ell_n + \Delta \leq j$, so either of the first two dispatching cases routes below j .

Thus every arrival decreases $H_{n,j}$ by exactly $1/n$. An external departure increases $H_{n,j}$ by $1/n$ precisely when its pre-departure pool occupancy is at most j . Phase progressions do not change $H_{n,j}$. Hence, on every time interval $[s, t]$ contained in a positive excursion,

$$H_{n,j}(t) - H_{n,j}(s) = \frac{C_{n,j}(t) - C_{n,j}(s)}{n} - \frac{A_n(t) - A_n(s)}{n}. \quad (5.13)$$

Lemma 3 and (5.12) imply, uniformly over such intervals,

$$H_{n,j}(t) - H_{n,j}(s) \leq -2\eta\varepsilon(t - s) + o(1). \quad (5.14)$$

If $j = 0$, then $H_{n,j} \equiv 0$, and (5.9) already gives $\ell_n = 0 = j$. Suppose henceforth that $j > 0$. Put

$$L = \frac{j+1}{2\eta\varepsilon}, \quad \sigma = \tau + L. \quad (5.15)$$

If $H_{n,j}(\tau) > 0$ and its current excursion did not hit zero before σ , then (5.14) would give

$$H_{n,j}(\sigma) \leq j - 2\eta\varepsilon L + o(1) = -1 + o(1),$$

which is impossible. Therefore $H_{n,j}$ hits zero no later than σ . Denote its first such time after τ by ζ_n ; if $H_{n,j}(\tau) = 0$, set $\zeta_n = \tau$.

We next examine the threshold at this capture time. If $H_{n,j}$ is already zero at the starting time, so that $\zeta_n = \tau$, then $\ell_n(\zeta_n) < j$ would imply

$$h_n(\zeta_n) = \ell_n(\zeta_n) + \Delta \leq j$$

and hence $q_n(\zeta_n, h_n(\zeta_n)) = 1$, contradicting (5.8). Thus $\ell_n(\zeta_n) = j$.

Otherwise, the last vacancy is filled by an arrival. Immediately before that arrival,

$$H_{n,j}(\zeta_n^-) = \frac{1}{n}.$$

There is therefore exactly one pool with $j - 1$ tasks, and all other pools have at least j tasks. In particular,

$$q_n(\zeta_n^-, j) = 1 - \frac{1}{n}.$$

If the pre-arrival threshold equals j , then (3.3) rules out a downward update, and $q_n(\zeta_n^-, j) < 1$ rules out an upward update; the threshold remains j .

If the pre-arrival threshold is below j , then its upper threshold is at most j . The invariant (5.8), together with the fact that every pool has at least $j - 1$ tasks, forces

$$h_n(\zeta_n^-) = j, \quad \ell_n(\zeta_n^-) = j - \Delta.$$

Furthermore,

$$q_n(\zeta_n^-, \ell_n(\zeta_n^-)) = 1, \quad q_n(\zeta_n^-, h_n(\zeta_n^-)) = 1 - \frac{1}{n}.$$

The upward-update condition in (3.1) therefore holds, and the post-dispatch threshold is again j . In all cases,

$$\ell_n(\zeta_n) = j. \quad (5.16)$$

Step 4: lock-in.

We have already shown that an upward update from j is impossible. Suppose, for a contradiction, that $\theta_n > \zeta_n$ is the first subsequent threshold-update time. It must be a downward update, so

$$q_n(\theta_n^-, j) \leq \alpha_n$$

and hence

$$H_{n,j}(\theta_n^-) \geq 1 - q_n(\theta_n^-, j) \geq 1 - \alpha_n. \quad (5.17)$$

Let β_n be the departure epoch that begins the positive vacancy excursion containing θ_n^- . This is the appropriate excursion endpoint: at β_n , a departure creates exactly one vacancy, so

$$H_{n,j}(\beta_n) = \frac{1}{n}.$$

Applying (5.13), (5.12), and Lemma 3 on $[\beta_n, \theta_n^-]$ yields

$$H_{n,j}(\theta_n^-) \leq \frac{1}{n} + o(n^{-\gamma_0}). \quad (5.18)$$

Together, (5.17) and (5.18) imply

$$n^{\gamma_0}(1 - \alpha_n) \leq n^{\gamma_0-1} + o(1) \longrightarrow 0,$$

contradicting (3.2). Thus no update occurs after capture. Since $\zeta_n \leq \sigma$, this proves (5.1).

Step 5: the stronger vacancy estimate.

Every positive excursion after ζ_n begins at height $1/n$. For any t in such an excursion, the same calculation gives, for every $\gamma < 1/2$,

$$H_{n,j}(t) \leq \frac{1}{n} + o(n^{-\gamma}),$$

uniformly over all excursions contained in $[\zeta_n, T]$. Therefore

$$\sup_{t \in [\sigma, T]} H_{n,j}(t) = o(n^{-\gamma}).$$

This proves (5.2). Since

$$0 \leq 1 - q_n(t, i) \leq H_{n,j}(t), \quad i \leq j,$$

(5.3) follows.

Step 6: upper-tail decay.

After time τ , no arrival creates a task beyond the first $j + \Delta$ slots of a pool. Indeed, when $H_{n,j} > 0$, arrivals are routed below j ; when $H_{n,j} = 0$, the invariant and (5.9) force $\ell_n = j$, and arrivals are routed below $j + \Delta$.

Consequently, Z_n can only decrease. An external departure from a pool with more than $j + \Delta$ tasks decreases Z_n by $1/n$. The normalized rate of these departures is

$$g_n(t) = \sum_{|\nu| > j + \Delta} \sum_{m=1}^r a_m \nu_m s_n(t, \nu),$$

and

$$g_n(t) \geq \eta \sum_{|\nu| > j + \Delta} |\nu| s_n(t, \nu) \geq \eta Z_n(t). \quad (5.19)$$

By Lemma 2, only finitely many profiles are active on each fixed horizon for all large n . The time-changed Poisson strong laws therefore imply relative compactness of Z_n . Along every convergent subsequence, any limit z satisfies

$$\dot{z}(t) \leq -\eta z(t) \quad \text{for almost every } t \geq \tau,$$

and thus

$$z(t) \leq z(\tau)e^{-\eta(t-\tau)} \leq u(\tau)e^{-\eta(t-\tau)}.$$

A subsequence contradiction upgrades this limit inequality to

$$\limsup_{n \rightarrow \infty} \sup_{t \in [\sigma, T]} Z_n(t)e^{\eta(t-\sigma)} \leq u(\tau)e^{-\eta(\sigma-\tau)} \leq u(\tau).$$

Hence (5.4) holds with the deterministic choice $c = u(\tau)$, which is independent of T . $\square$

Corollary 1. *Under the refined Coxian scheme, the compact-horizon conclusions of the basic-scheme Coxian theorem remain valid, and the lower-occupancy conclusion strengthens to*

$$\sup_{t \in [\sigma, T]} \sum_{i=1}^j (1 - q_n(t, i)) = o(n^{-\gamma}) \quad \text{for every } \gamma < 1/2.$$

6 Why stabilization is not uniform over the infinite time horizon

The preceding theorem does not say that a fixed sufficiently large system eventually keeps the same threshold forever. In fact, the opposite is true.

Proposition 1 (Infinite threshold recurrence at fixed n). *Fix $n < \infty$ and $\alpha_n \in (0, 1)$. Under the refined learning scheme, the threshold changes infinitely often almost surely.*

Proof. Let $K_{n,m}(t)$ denote the total number of phase- m tasks in the entire system. The aggregate process

$$K_n(t) = (K_{n,1}(t), \dots, K_{n,r}(t))$$

does not depend on the routing or on the threshold. It is an open infinite-server Jackson network whose stationary phase counts are independent Poisson random variables with means

$$\frac{n\lambda r_m}{\mu_m}, \quad 1 \leq m \leq r.$$

Its stationary empty-state probability is therefore

$$\prod_{m=1}^r \exp\left(-\frac{n\lambda r_m}{\mu_m}\right) = e^{-n\rho} > 0.$$

The aggregate chain is irreducible and positive recurrent, so it returns to the empty state infinitely often almost surely.

Suppose that the threshold changed only finitely often. It would then eventually equal a constant $L \in \Delta\mathbb{N}$.

If $L > 0$, consider a return to the empty state after the alleged final threshold update. Immediately before the first subsequent arrival,

$$q_n(L) = 0 \leq \alpha_n.$$

That arrival therefore triggers a downward update, a contradiction.

It remains to consider $L = 0$. Starting from an empty system with threshold zero, there is positive probability that the next $n\Delta$ events involving the tasks are all arrivals, with no intervening phase completion. For example, with $\mu_{\max} = \max_m \mu_m$, this probability is bounded below by the strictly positive number

$$\prod_{k=0}^{n\Delta-1} \frac{n\lambda}{n\lambda + k\mu_{\max}}.$$

During such a burst, every task is routed to a pool containing fewer than Δ tasks unless an upward update has already occurred. If no earlier upward update occurs, then immediately before the $n\Delta$ -th arrival, $n - 1$ pools contain at least Δ tasks and the remaining pool contains $\Delta - 1$. Hence

$$q_n(0) = 1, \quad q_n(\Delta) = 1 - \frac{1}{n},$$

so the $n\Delta$ -th arrival triggers an upward update.

If the threshold were to remain zero, the system would visit the empty state infinitely often. By the strong Markov property, each visit gives another trial with the same strictly positive success probability. Almost surely, one trial eventually succeeds, contradicting permanence of the zero threshold. Thus the threshold cannot become constant and must change infinitely often. $\square$

Remark (Order of quantifiers). Theorem 1 asserts that, on a probability-one event,

$$\forall T > \sigma \quad \exists N_T(\omega) \quad \forall n \geq N_T(\omega) \quad \forall t \in [\sigma, T] : \ell_n(t) = j.$$

It does not assert

$$\exists N(\omega) \quad \forall n \geq N(\omega) \quad \forall t \geq \sigma : \ell_n(t) = j.$$

The latter assertion would contradict Proposition 1. There is therefore no conflict between compact-horizon many-system stabilization and infinite threshold recurrence in every fixed finite system.

References

- [1] Diego Goldsztajn, Sem Borst, Johan S. H. van Leeuwaarden. *Learning and Balancing Unknown Loads in Large-Scale Systems*. Mathematics of Operations Research, 2024. <https://doi.org/10.1287/moor.2021.0212>
- [2] D. Goldsztajn, S. C. Borst, and J. S. H. van Leeuwaarden, *Learning and balancing unknown loads in large-scale systems*, arXiv:2012.10142v3, 2024. <https://arxiv.org/abs/2012.10142>.